

Zero-Trust Architectures and Wireless Edge Security

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***Abstract*—Traditional perimeter-based wireless security models cannot adequately protect modern enterprises that rely on mobile devices, Internet of Things (IoT), cloud services, and distributed edge computing, as they expand attack surfaces through multi-tenancy and disappearing perimeters in 5G environments. Zero trust architecture (ZTA) replaces implicit trust based on network location with continuous verification of users, devices, and traffic—"never trust, always verify"—making it well suited to securing wireless access networks and edge environments where devices exhibit high mobility and heterogeneity. This paper analyzes how zero trust principles, as defined by NIST SP 800-207, can be applied to enterprise Wi-Fi and 5G edge deployments to mitigate threats such as rogue access points, lateral movement from compromised IoT nodes, and multi-tenant vulnerabilities in edge computing. The methodology combines a literature review of standards like NIST SP 800-207 and recent 5G research with a conceptual design of a zero trust-enabled wireless edge, focusing on identity-centric access control, device posture assessment, micro-segmentation, and continuous monitoring.**

Findings indicate that integrating ZTA with wireless and edge infrastructures reduces attack surface, improves detection and response through equipment ID-based isolation, and provides granular control over access to critical resources, although it introduces complexity, interoperability challenges with legacy devices, and operational overhead in multi-cloud 5G scenarios. The paper concludes with practical guidance and assumptions for organizations planning to evolve from perimeter-centric wireless security to a zero-trust model at the edge.

I. INTRODUCTION

Enterprise wireless networks have evolved from convenient add-ons to critical infrastructure that supports mobile workforces, IoT devices, and latency-sensitive applications at the network edge, particularly as 5G enables Multi-Access Edge Computing (MEC) and private networks. As Wi-Fi and cellular connectivity extend deeply into operational environments such as campuses, factories, and hospitals, the traditional security assumption that devices inside the perimeter are inherently trustworthy has become untenable, especially with BYOD policies, hybrid work, and

multi-tenant edge platforms increasing vulnerabilities to lateral movement and exploits like Log4j.

Zero trust architecture responds to this shift by abandoning location-based trust and focusing instead on users, assets, and resources, with NIST SP 800-207 defining zero trust as cybersecurity paradigms that reduce uncertainty in enforcing least-privilege access through continuous evaluation of identity, device posture, and context. When applied to wireless and edge environments, zero trust can constrain lateral movement from compromised wireless clients, enforce granular access to applications via network slicing and ZTNA, and maintain security even as devices roam and attach from untrusted networks in 5G industrial systems.

This paper concentrates on how zero trust principles can be mapped into practical controls and architectures for securing enterprise wireless and edge deployments, particularly Wi-Fi and private 5G environments, drawing on NIST guidance and 5G-specific research. The analysis focuses on design patterns and control mechanisms—such as visibility, ML-based threat prevention, and equipment ID enforcement—rather than vendor-specific products, with emphasis on identity, segmentation, and monitoring at the wireless edge

II. SCOPE

The scope of this work is limited to the application of zero trust to enterprise wireless access networks and edge computing platforms, including Wi-Fi and private 5G deployments that connect users and devices to local and cloud-hosted services amid

rising multi-tenancy risks. Public carrier networks, consumer-grade Wi-Fi, and full enterprise-wide zero trust transformations are outside the primary scope, except where they inform edge and wireless security patterns in MEC and industrial IoT contexts.

The paper focuses on the control plane and policy aspects of ZTA—identity, authentication, authorization, segmentation, and monitoring—as they relate to wireless clients and edge workloads, aligned with NIST SP 800-207's logical components. Lower-level radio-frequency topics such as antenna design, modulation schemes, or detailed 3GPP protocol analysis are treated only at a high level where they intersect with access control and security policies in 5G zero trust synergy.

III. OBJECTIVES

The goal of this research report is to explore, assess, and gain understanding of the implementation of Zero-Trust Architectures and Wireless Edge Security, to ultimately identify effective strategies and frameworks that strengthen data protection, network security, and trust management in distributed, mobile, and wireless environments.

This research report will serve the purpose of analyzing Zero-trust Architectures and Wireless Edge Security through developing components that explains how these types of technologies are used. Various components developed are detailed logical diagrams, physical diagrams, architectural details, functional requirements, security solutions and more. This report will determine correlation between fundamental security concepts and implementation

of these technologies. We will also compare the effectiveness of Zero-Trust Architectures versus Wireless Edge Security practices. Lastly, we will identify key challenges in implementing these technologies effectively from real-world settings for companies and organizations.

IV. METHODOLOGY

The following study adopts a desk-based research design with a structured review of online literature on Zero-Trust Architectures and Wireless Edge Security. The purpose to collect and understand concepts, architectures, and best practices rather than collect or survey new experimental data.

The main source categories we collect and focus our online research on are peer-reviewed articles, official standards and guidelines, technical blogs, and security reports. Peer-reviewed articles will be retrieved from digital libraries such as IEEE Xplore and open-access repositories, standards and guidelines will include NIST publications and industry frameworks, and only reputable cybersecurity and networking organizations will be considered for blogs and security reports.

For each source selected, important information will be extracted into a notes document that includes definitions, components, security principles, threat models, and control mechanisms that use Zero-Trust Architecture and Wireless Edge Security. This report will show our analysis of comparing and integrating our findings across sources, identifying patterns, ideas, and disagreements in how these technologies are applied. Concepts and controls that consistently appear across publications will be treated as

foundations for an understanding of Zero-Trust-based Wireless Edge Security.

V. RESEARCH DETAILS

A. Zero-Trust Architecture

The following sections includes an overview of concepts and ideas of Zero-Trust Architectures and Wireless Edge Security that address how to protect modern and vastly interconnected networks.

From our research, we learned that Zero-Trust Architectures (ZTA) is an important model of security from the foundation of “never trust, always verify”. Meaning, there is no user, device, or service that will be trusted by default, whether from an internal or external source from the network. Strong identity verification, least-privilege access, continuous authentication, and policy-based access decisions are the relevant core security principles that enforce the protection of sensitive resources. ZTA relies on identity and access management, segmentation, gateways, and continuous monitoring to reduce trust and prevent lateral movement by attackers. Standards and regulation bodies such as NIST provides references and guidance that organizations or companies can use as templates for implementing a ZTA.

Our research shows that with the increase of sophisticated threats, supply chain risks, APIs, and microservices that connect directly to the internet, implied trust can be dangerous. Assuming that all inside a network is trustworthy will be collapsed by cloud services, remote work, and distributed applications. Organizations with hybrid cloud, on-premise, and mobile environments can better protect

sensitive data by tying access to identity, device posture, and context, while continuously reassessing risk during each session.

B. Wireless Edge Security

Our research on Wireless Edge Security provided us the understanding the security concepts related to devices and services operating near end users. 5G edge nodes, IoT sensors, industrial controllers, and smartphones are edge devices that connect from wireless technologies. Wireless edge environments often have small, low-power electronics, unstable connectivity, and large numbers of endpoints deployed externally. Data is processed close to where wireless edge devices are deployed to reduce latency and bandwidth use. The distribution of edge devices increases the attack surface compared to centralized networks.

Threats to wireless edge devices may include device compromise, rogue access points, eavesdropping, and exploiting outdated protocols. These threats are difficult to prevent because many devices are physically exposed, difficult to patch, and communication traverses over public radio channels. Resource-constrained devices may especially not have sufficient security controls as they require lightweight authentication, encryption, and anomaly detection techniques.

C. Applying Zero-Trust Architecture to Wireless Edge Environments

Treating all connections and devices as untrusted until verified is an increasingly proposed Zero-Trust framework to secure 5G+ networks, edge clouds, and IoT deployments. In many cases, ZTA applies to

policies at edge gateways, base stations, and service meshes while enforcing strict identity verification, device authentication, and traffic segmentation. ZTA and Wireless Edge security standardization aim to define applicable and appropriate approaches so operators can consistently implement Zero Trust at the edge layers. This motive comes from the need to handle large, decentralized wireless edge environments with software-defined networking and anomaly detecting technology that combines with ZTA.

In our research, wireless edge security is often an important application for Zero-Trust Architectures, especially in edge computing environments. The intersection of these technologies proves an active research area for creating a layered-security solution that designs resilient and adaptive defenses against emerging and evolving cyber threats.

VI. RELEVANT FINDINGS

When it comes to the relevant findings for Zero Trust Architecture and Wireless Edge Security, some strong points will be broken down. Firstly when it comes to zero trust architecture, it can be applied to various applications. According to ((IEEE ComSoc, Paragraph 2) ZTA can protect against unauthorized access, and its deployment is starting to become an industry standard. Methods that can be used include network segmentation to enhance security, multi-factor authentication, even autonomous monitoring to ensure that devices and network traffic are functioning as they should.

Other methods of network security also include the use of Wireless Lan Controllers throughout the

network to monitor and control the available access points, as this will also help in boosting the available security within the network along with help to enforce ZTA.

Other findings also find the specific vendors also provide effective ZTA solutions to clients, such as Fortinet. According to (Fortinet, *What is Zero trust edge? What differentiates Fortinet's Zero Trust Edge architecture?*) Fortinet offers multiple ZTA security options. This includes options like Industry leading SD-Wan options, where they enable organizations to scale at any level. They also have access to Hybrid Convergence, which allows for a management console powered by FortiOS to handle traffic inspection and network wide monitoring for both wired and wireless devices. And finally, there is also ZTNA Everywhere, where they offer ZTA proxy options that are natively enabled via FortiGate appliances along with their virtual machines. These can all be activated on or off the premises.

Overall, from the relevant findings, it can be seen that ZTA solutions can not only be followed with dedicated guides on the process but also includes vendors which offer ZTA options which also accommodate wireless edge solutions.

VII. SUPPORTING DETAILS

As defined by NIST SP 800-27, zero trust architecture models are becoming industry standard due to its focus on protecting network infrastructure and sensitive data through its policy of not trusting the source on its face and choosing to verify the source before allowing access.

WLAN controllers like those offered by Fortinet provide a centralized approach to monitoring a ZTA-based network infrastructure through its policy of “All SSIDs are created as interfaces on the FortiGate, so all traffic is VLAN/subnet isolated without the need to actually deploy those VLANs or even map to existing VLANs; they exist in the context of the controller” [Fortinet, 1].

VIII. CHALLENGES

Some of the challenges regarding the use and implementation of zero trust architecture with wireless edge security are facing operational challenges, dealing with compatibility issues, and challenges regarding employees resisting these changes and models.

Operational challenges can be a rather large factor when it comes to utilising zero trust architecture due to its drawback of all employees requiring access to various systems and information as needed in order to perform their daily tasks and requirements. This can lead to slower and reduced efficiency as these employees need to continuously request access, interrupting workflows[terranovasecurity, 2].

Compatibility issues may also exist in environments with a complex infrastructure, as each endpoint and service will need to be fitted with ZTA and still be able to interface with employees and users while remaining secure [fortinet, 3]. This challenge could lead to significant security vulnerabilities in the event of a service not being correctly configured with ZTA principles, allowing the employees and users to access information that

they do not necessarily have the proper clearance to view or edit.

Employees may also resist these new changes to systems, not liking or wanting to deal with a heightened security presence and preferring the pre-existing systems as they are [Ankura, 4]. A method to combat this resistance and hesitation would be to introduce new training for how to interface with ZTA and directing managers to enforce these new changes.

IX. ASSUMPTIONS

Some assumptions that we made when analyzing the scope and conducting our research are that we are working with an unlimited budget to purchase network devices with and that there is no RF interference from neighbouring devices that have the potential to cause issues. An unlimited budget is necessary for our research because the focus of this paper is not to come up with a financially viable methodology but a theoretical best possible scenario. A limitation of this is that for many enterprises elements of this research may not work for them due to financial restraints in which case this paper should be used as a starting place to give guidelines on edge security.

It is important for this research that there is no RF interference from neighbouring devices being emitted by other enterprises in an adjacent building for example. Although it is possible that in a real-life, practical situation this is possible, we are assuming that this network exists in a vacuum and free from this type of interference.

X. ADDITIONAL REPORT SECTIONS (COMPLICATIONS)

When it comes to additional report sections, it can also be discussed that ZTA and Wireless Edge Solutions can bring on complications with security, especially with the difference among device operating systems and how they function. Issues with the concept of BYOD (bring your own device) also play into this issue, as with ZTA it would be more effective for company issued devices to be given, rather than an individual having to have their personal device (whether apple or android), constantly monitored on the company network along with possible doing this off-site.

Aside from operating system differences, and BYOD situations, another complication that would be relevant to ZTA would be legacy hardware. If the devices being monitored are long outdated with the current security standards being used, whether a user brings it to the office, or it was company provided, the ZTA state of monitoring will already be at risk. If the device is far too old for modern security standards, not amount of ZTA will keep safety in check, unless the device is not exposed to the internet or brought into the company network ecosystem.

One final issue may also be with network scaling and integrating with existing infrastructure. ZTA solutions often require more processing power, and protocol usage. This can play a role in hindering the capability for a company to scale their network up, along with making use of devices that are considered legacy devices in the company network. So while ZTA and Wireless Edge Security are powerful tools, it does come with relevant cons that may persuade a

company not to make use of it, depending on their security situation and goals.

XI. CONCLUSION

In conclusion when it comes to ZTA and wireless edge security, there are many angles that need to be considered before choosing to implement it. Whether choosing a specific vendor, checking for device compatibility, or unrolling company provided devices, it is most important that the company or user understands the security goals they have, and why ZTA may be the chosen option. While ZTA is powerful, it may not be the solution that is best for all company environments. Above all else, ZTA cannot be used as a crutch to hold up the company's security solutions, as the first line of defense aside from network architecture, are the people that are trained to use it. No amount of architecture can prepare or protect a company against poor employee security training.

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